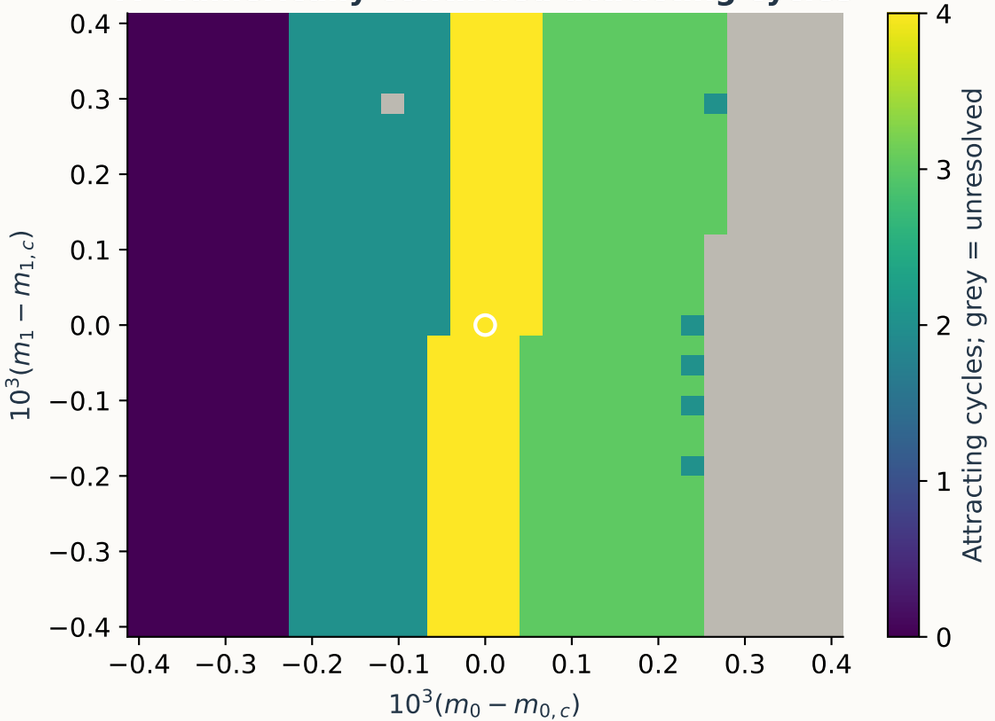
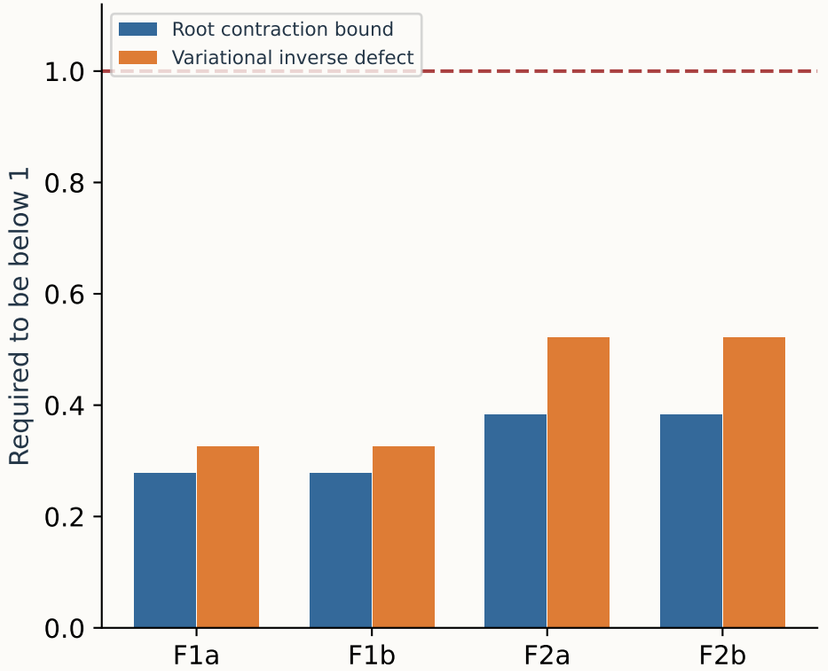
# Four coexisting periodic orbits persist on a parameter box

## A Numerically continued attracting cycles




## B Uniform interval tests for all four cycles




Certified simultaneously

$|\alpha - 9.06| \le 10^{-8}$
$|\beta - 14.3458| \le 10^{-8}$
$|m_0 - (-8/7 + 0.006)| \le 10^{-8}$
$|m_1 + 5/7| \le 10^{-8}$
$0 \le \gamma \le 10^{-8}$

The certified box is much smaller than one cell of the numerical map. It is a conservative proof region, not the measured stability boundary.